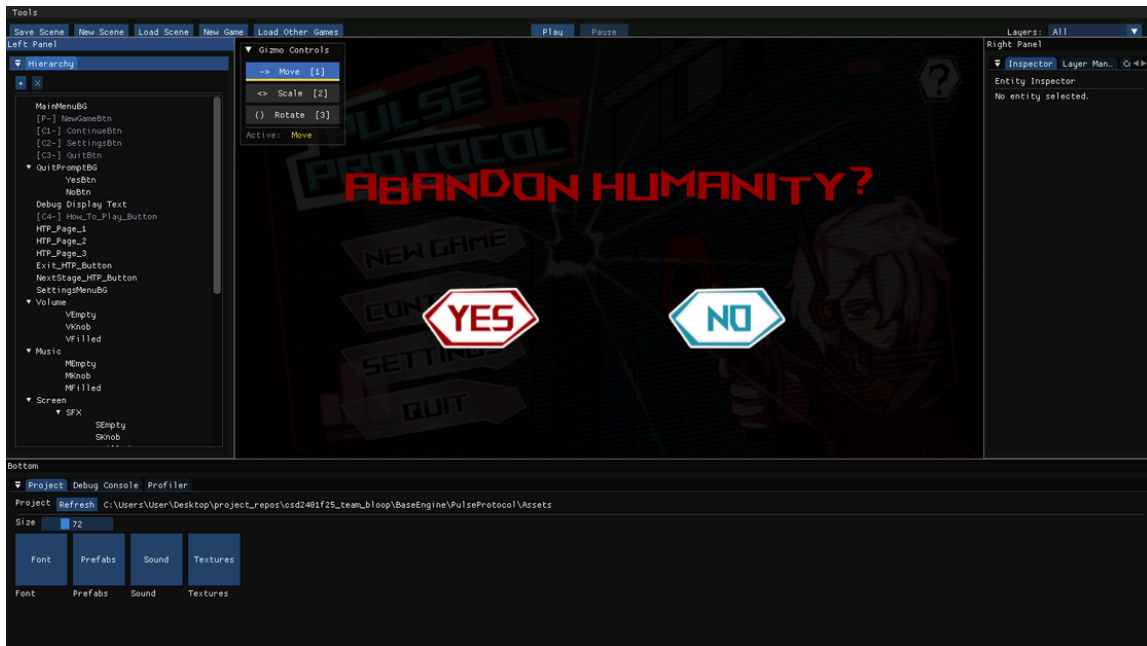


PulseEditor Guide

Editor Overview



PulseEditor is a level editor for *Pulse Protocol*. It provides tools to build levels, place entities (heroes and enemies), configure game objects, control visibility of layers and manage project assets. The editor interface consists of seven main sections:

- A. Toolbar** - Quick access to scene management (Save Scene/ New Scene/Load Scene/New Game/Load Other Games) and playback controls (Play/Pause) + Build Size Analyzer.
- B. Hierarchy Panel** - View and organize all entities in your current scene.
- C. Viewport** - Visual representation of your level with real-time rendering.
- D. Inspector Panel** - View and modify properties of selected entities including transforms, textures, animations, and rendering options
- E. Project Browser** - Access and manage all game assets organized by type (Font, Music, Prefabs, SFX, Sound, Textures). Includes Debug Console and Profiler tabs for testing.
- F. Layer Manager** - Allows users to control the visibility of each layers. Currently there are “Background”, “World” and “UI” layers.
- G. Config Manager** - Allows users to view and edit core game configuration settings directly from the editor, without manually editing any files.

See the following pages for detailed explanations of each section.

A. Toolbar



Save Scene Load Scene

Save Scene - Opens a file browser to choose where to save the current scene. Saves the level to a JSON file by serializing all entities with their complete component data (position, scale, rotation, textures, animations, etc.). This preserves your level design for later use.

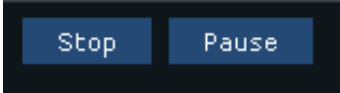
Load Scene - Opens a file browser to select a previously saved scene JSON file. Loads the selected scene by recreating all entities exactly as they were saved. When loading a new scene, all entities from the previous scene are automatically removed first.

Playback Controls (3 modes shown):



Play

Pause



Stop

Pause



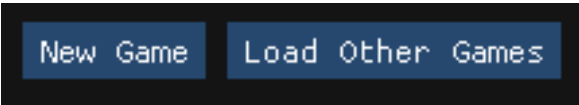
Stop

Resume

Mode 1: Play / Pause - Play: Start game simulation in editor - Pause: Pause game execution

Mode 2: Stop / Pause - Stop: End game simulation and return to edit mode - Pause: Pause game execution

Mode 3: Stop / Resume - Stop: End game simulation - Resume: Continue from paused state



New Game

Load Other Games



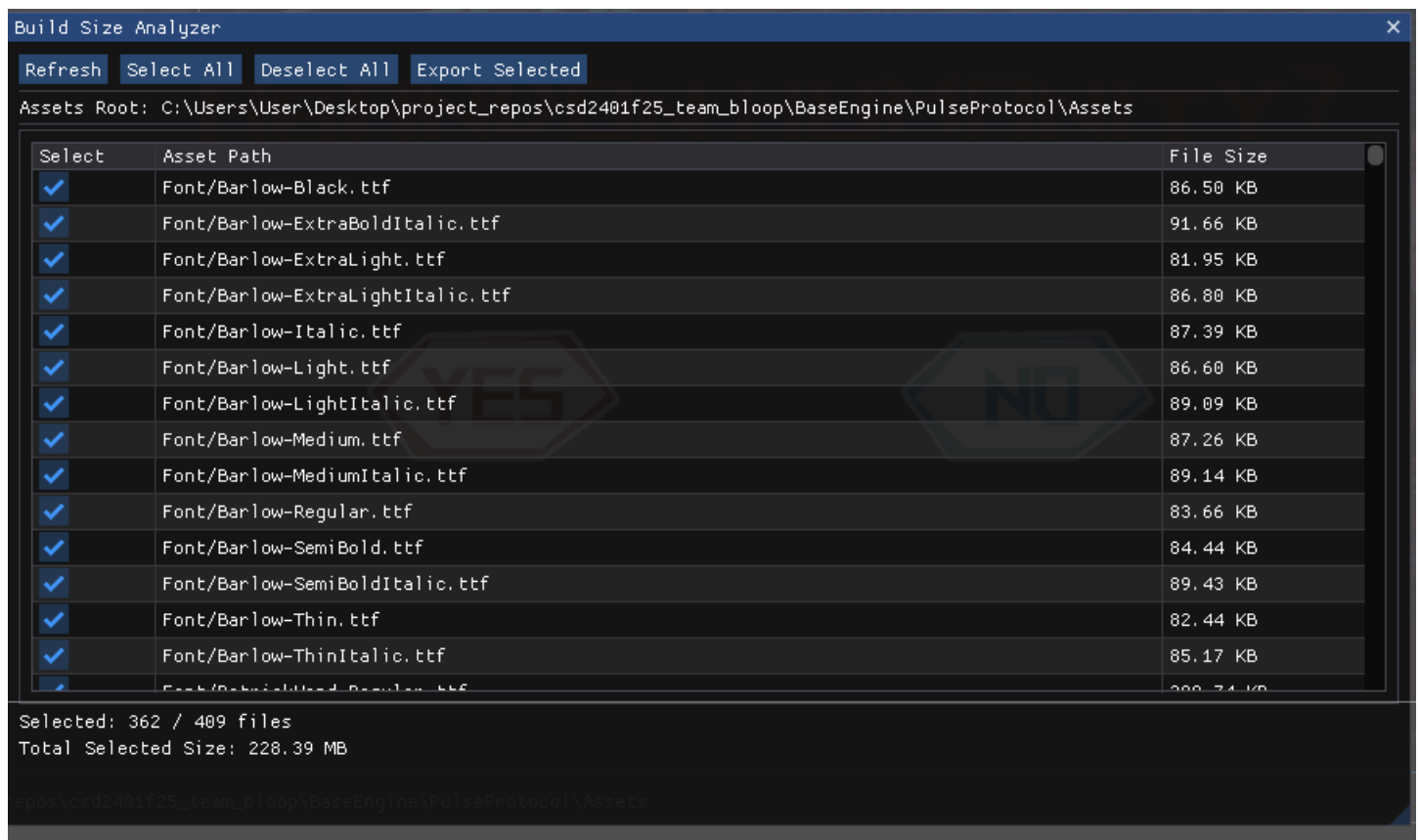
New Scene

New Game - Opens a popup to enter a name for a Scene.

New Game - Opens a popup to enter a name for a new game project. Creates the full folder structure (assets, JSON files, ManagedScripts C# project) under BaseEngine. *Requires an editor restart to begin editing.*

Load Other Games - Opens a file browser to select a config.json from an existing game project. Switches the active game to the selected project. *Requires an editor restart to take effect.*

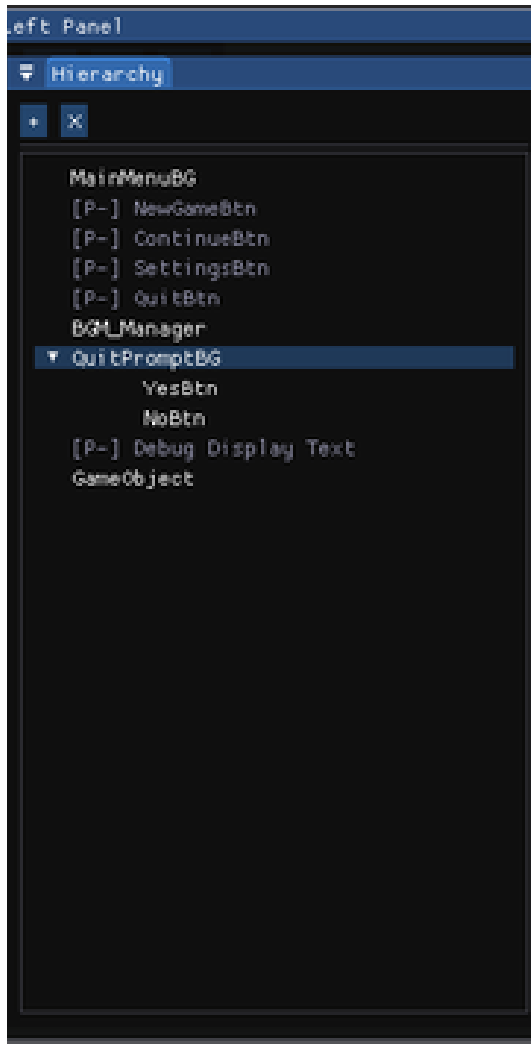
Located in the “Tools” button, there is a build size analyzer.



Functions:

- Scans your Assets folder and lists every file with its size
- Lets you check/uncheck which assets to include
- Has Select All / Deselect All
- Actually copies the selected files to a destination folder via "Export Selected"
- Saves your selection state between sessions

B. Hierarchy Panel



The Hierarchy Panel displays all entities currently in your scene.

This is where you select, organize, and manage game objects.

Features:

- **“+” adds a Empty GameObject by default**
- **“X” button removes/deletes an entity**
- **Able to group entities together**

Click any entity name to select it. Selected entities are highlighted and their properties appear in the Inspector panel (Section D).

C. Viewport



The central workspace for visualizing and interacting with your game scene.

What You Can Do:

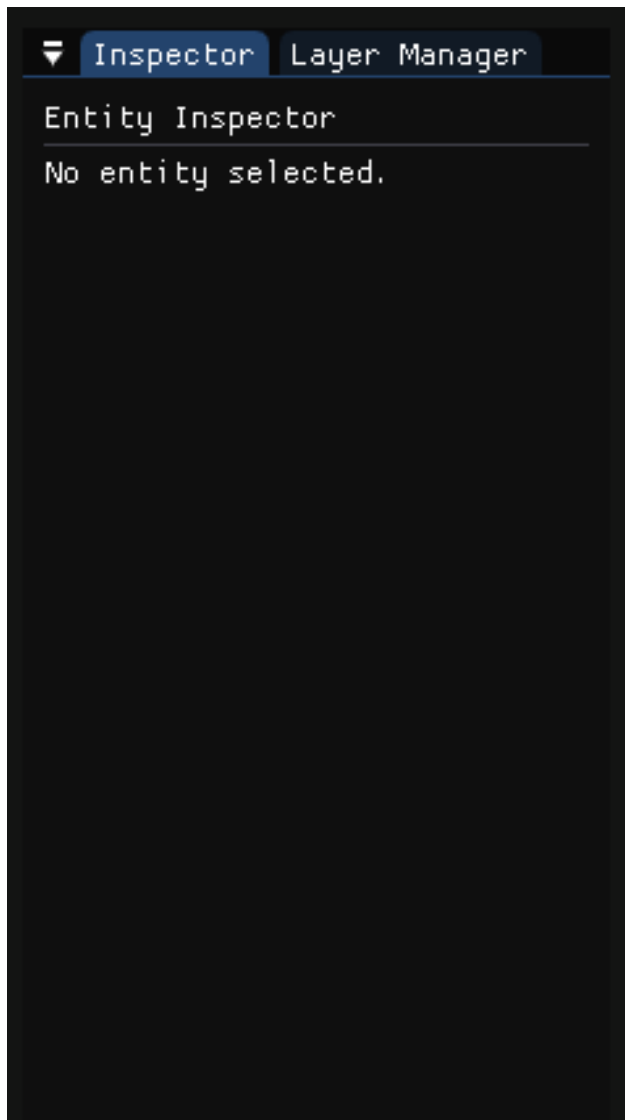
- **Select Entities** - Click any object to select (yellow border appears)
- **Group Entites together** - Drag an entity to any existing entity to group it together
- **Test Gameplay** - Use Play/Pause/Stop buttons to test your level
- **Zoom In/Out** - Scroll mouse wheel to zoom in or out of the scene
- **Deselect** - Click empty space to clear selection

Modes (Using the toolbar):

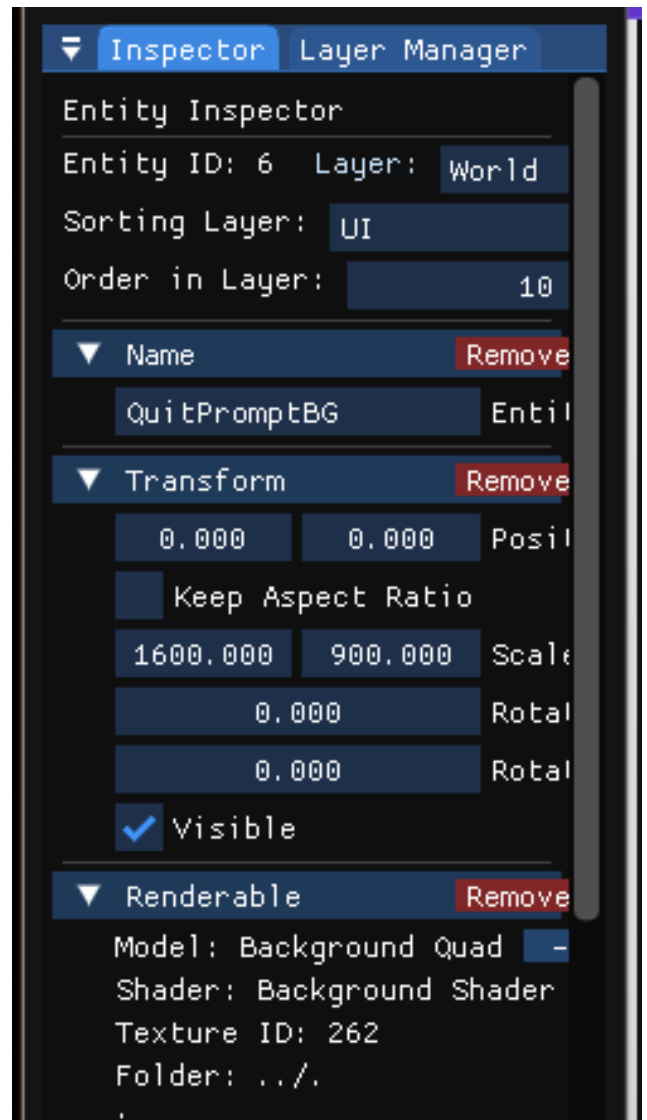
- **Edit Mode** - Freely modify entity positions and properties
- **Play Mode** - Test gameplay mechanics and interactions
- **Pause Mode** - Freeze game state for inspection
- **Stop Mode** - Restore scene to original state

Note: Changes during Play mode are temporary and reset on Stop!

D. Inspector Panel



Screenshot 1:
No entites selected.



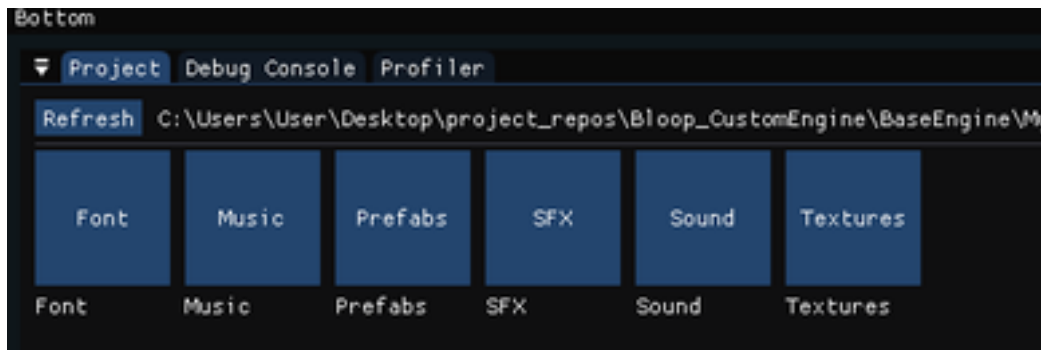
Screenshot 2:
QuitPromptBG Entity selected.

The Inspector displays detailed properties of the currently selected entity. To view properties, **select an entity from the Hierarchy panel first.**

Usage:

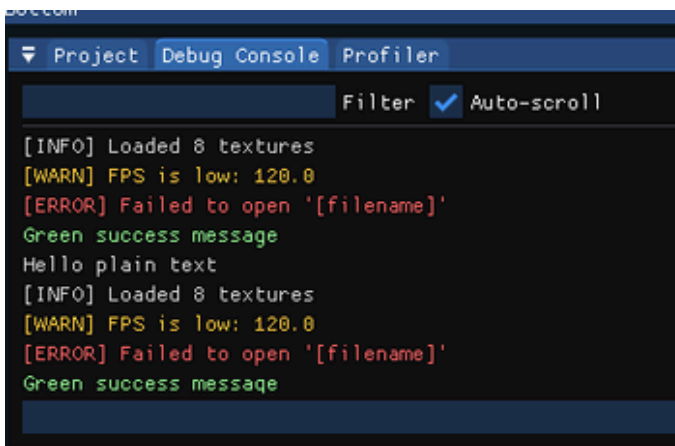
The viewport shows the current state of your level. Any changes made in the Inspector or Hierarchy update in real-time here. Click the Play button in the toolbar to test your level in-game and see how all changes work during gameplay.

E. Project Browser



The Project Browser organizes all game assets by type, making it easy to find and use resources in your level.

Usage: Click tabs to browse asset types. Drag and drop assets into the Hierarchy panel to quickly add them to your scene. Prefabs are especially useful for placing pre-configured entities (enemies, props, etc.) without manual setup. Use Refresh after adding new files externally.



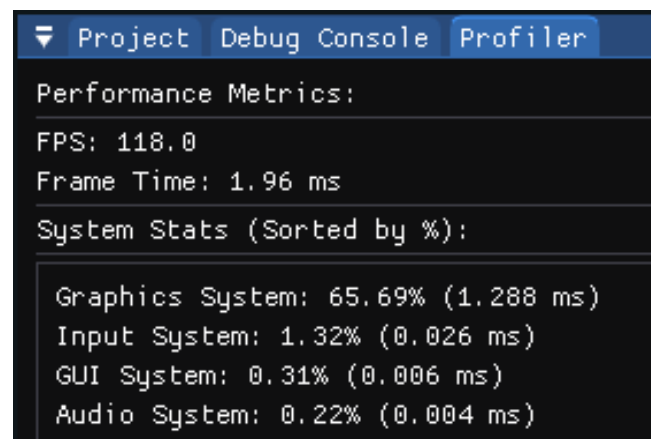
Displays runtime messages, warnings, and errors during testing (currently in development).

Features:

- Real-time logging of system messages
- Error tracking for troubleshooting
- Warning notifications for potential issues

Usage:

Monitor this tab while testing your level in Play mode to catch bugs, track game events, and identify issues that need fixing.



Provides performance monitoring and optimization data for your game.

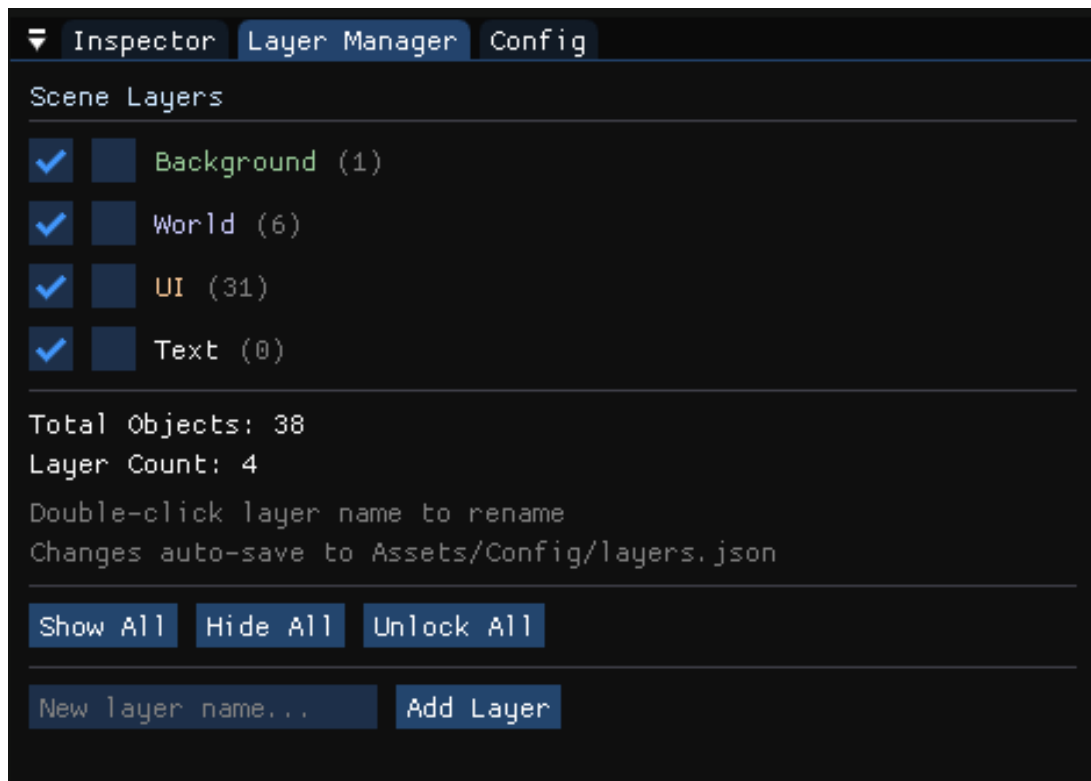
Features:

- Frame rate (FPS) tracking
- Graphics system load percentage
- Performance metrics for identifying bottlenecks

Usage:

Use the Profiler to monitor game performance during testing. Check for frame rate drops or high system load that might indicate optimization needs. Helps ensure your level runs smoothly on target hardware.

F. Layer Manager



The layer manager allows you to organize, control visibility, and manage the rendering order of entities across different visual layers of your scene.

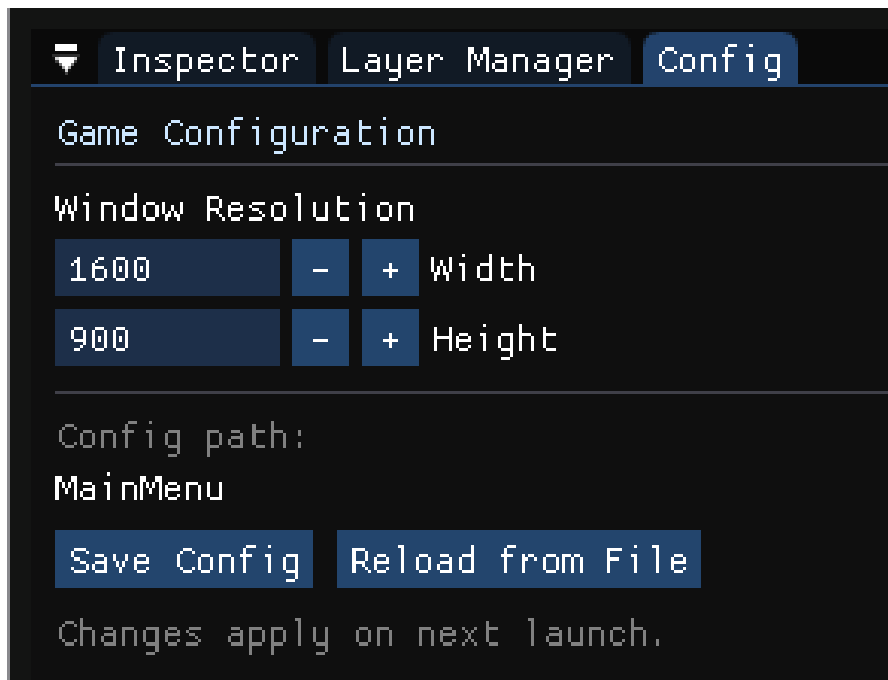
Controls:

- **Show All** - Makes all layers visible at once. Useful when you need to see your complete scene composition.
- **Hide All** - Hides all layers simultaneously. Use this when you need a clean viewport for focused work or screenshots.
- **Unlock All** - Unlocks all layers, allowing editing on every layer. Convenient when you want to freely manipulate any entity.
- **Add Layer** - Create new custom layers beyond the default three. Enter a name in the text field and click "Add Layer" to create a new organizational category.

Usage:

It shows the current organization of all entities in your scene. Any changes made in the Inspector or Hierarchy update in real-time here. Click the Play button in the toolbar to test your scene in-game and see how layers and their visibility settings work during gameplay.

G. Config Manager



The config manager allows you to view and edit core game configuration settings directly from the editor, without manually editing any files.

Controls:

- **Save Config** - Writes all current field values to *config.json* in the source directory. A confirmation message appears on successful save.
- **Reload from File** - Reverts all fields back to the currently saved values in *config.json*, discarding any unsaved changes.

Usage:

It shows the current configuration values loaded from *config.json*. Any changes made to the fields only take effect after clicking **Save Config**. Click the **Save Config** button after editing to persist your changes, then relaunch the game to apply them. Use **Reload from File** at any time to discard edits and restore the last saved state.